

Bradly Alicea: “An Overview of Open Source and Open Science”

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all right the next recorded talk the penultimate recorded talk is by Bradley Alysia earlier Bradley gave a talk on purposefully non-standard cybernetics and this talk is going to be an overview of Open Source and open science so that's really cool and thanks for bringing this important complimentary topic to the Symposium for all all all who are working on this okay it's 51 minutes long hello my name is Dr Bradley Alisa and I'm a member of the active inference Institute scientific Advisory Board today I'm going to give you an overview of Open Source and open science I'm my home institution is the orthogonal research and education lab I'm also affiliated with the openworm foundation and the University of Illinois champagne or champagne I'm also with particular interest in the orthogonal research and education lab is our open source interest group and I'll talk about that throughout this talk this starts with a question what can we do with open source and open SS and I pair these together because sometimes we're building software sometimes we're doing scientific discovery which kind of relies on the open source software and so if we're doing both we can actually build uh systems that can sort of help us do better software engineering and science so they're kind of grouped together so in terms of Open Source you know it it's maybe a truism that open source is about software and putting it out into the world but that's not enough and you really want to what you want to do instead of just putting software into the world is building a community around technical standards and code and this is true if you want to build a community of users or an open source community that helps build the software because you need to have some interaction with the software to make it sustainable you also want to have good technical standards and with academic software as we see that's been lacking this also allows us to open up the organization's processes and products so having these communities and having these technical standards are not only good from just just stamping of sharing software but it also opens up our processes and products and allows us to serve our communities better our academic community ities open science is where we share data share results share research

artifacts of the public so we do a lot of sort of transparency related things we give people our results and our data and it's software and papers and presentations and we're very open about we can give them access to these things so when I say open science I also kind of mean Open Access and it's they're kind of in the same bucket open science in general allows us to open up our organization's research capabilities so that people can benefit from our discoveries and we can benefit from their feedback so a lot of organizations have propose different sort of road maps or statements on open science and what that involves so not every organization's view of this is going to be the same but I'm using the young European Arch University's Network statement on open science at the link below and to them you know they have a very strong emphasis on collaboration and citizen science so this is where you have collaboration around the world around the scientific community and also involving say citizen science where people just want to contribute their own um analyses sometimes it's data sometimes it's observations things like that there's visibility and idea sharing so know being able to get to the outputs and read them and understand them is important open education is important idea sharing because you need to have you know uh focus on collaboration alternative models of publishing you need to be able to publish the work to get it out there into people's hands incentives assessments and research Integrity so those are all kind of part of this open science tree now again every organization who Champions open science will have a different view of this we don't really have a formalized view in the orthogonal lab but it follows this kind of outline close quite closely and so if you want to know more about the where the open or where the orthogonal research lab is coming from you can read this paper and I put this out in 2020 he needs to be updated but it's a good overview and this is building and distributed virtual Laboratory adjacent to Academia um and and so there are a number of things in this paper it's just basically it's it's very similar to the model that the active inference lab uses were much more focused on sort of educational opportunities and opportunities for people to sort of pick up the load and uh build uh research artifacts and you know having students come in and do short-term projects but it's it's it's a very similar process so one of the processes or one of our goals is to integrate open access in our open source practices for advancing distributed research projects so we want to be able to help people make these things open help them build research projects the other thing is that we want to encourage practice on open working through the creation of digital research artifacts it's like papers or presentations and when I say open working I mean I borrow this term from uh the Mozilla foundation and they're approach to this whole Enterprise is basically working open so it's working in these collaborative

distributed groups and how to make the most of that kind of experience you know things like note taking and meetings and managing sort of the human capital so that's also part of it open source and open Sciences managing human capital managing these work groups so one of the big thing other big things about open source and open science this idea practice and so you know it's it's very practice driven as opposed to sort of theoretical or philosophically driven and so you know practice means that you need to establish practices procedures routines workflows and Technical Solutions for working open so this means you need to adopt open practices it could be Community review of code releasing code under a license printing manuscripts and generating open data sets so these are all things that involve you know planning but also practice and being able to produce something at the end related to that are what we call open workflows so when we want to release an open data set we can't just pop data out onto the internet we have to put open data sets into a production line model where we you know clean the data we analyze the data we produce metadata and then we publish that formally in a repository with metadata attached we also want to have Community standards for open participation so we can't just expect people to pick things up and run with them without giving them some standards for working so you know you know usually in academic work you're trained a certain number of years it becomes kind of intuitive to you what to do but a lot of people especially students undergraduates they come in and they don't know what to do they don't have like those standards ingrained and so you have to have Community standards but also because sometimes the incentives aren't there for people to follow long lengthy procedures and so they don't in any case this helps you standardize this e these efforts then we have open Technical Solutions so open source and open science are built or enabled by technical solu Solutions you couldn't really have open source and open science without having open Technical Solutions like GitHub live streaming tools and bibliographic tools and of course those things don't make something open source or open science by themselves but they enable those practices so this is a nice graph where I kind of talk about vision of sustainable open science and so I have a couple of categories here to the left and the right which are are sort of suboptimal areas and then where we want to go is we want to go to this Zone this elusive Zone in the red box so the elusive Zone the red box are the set of tools that would allow us to do sustainable open science and Open Access and the stuff to the left and the right are kind of like things that we do now that are suboptimal so we start with the suboptimal and we talk about black Open Access which is kind of an fancy word for piracy or informal sharing and so each of these types of open aess are colored black just means it's kind

of related to under the table or you know piracy or something so one of the examples are is syab where you know it's I my position on piracy is it's Justified given the incentive structure in science but you know you have systems where you can just get access to things and then of course sometimes we share PDFs by sending emails to our colleagues and sending them the paper and that's all black Open Access because it's very informal and it's not really sustainable because you're relying on people to have you know a you know someone has to get access to these and then share it with someone else um then you go over to the other side you have two other areas of Open Access you have gold Open Access and diamond platinum Open Access so gold open and access is where typically an author will pay APC fee which is a publication fee it's usually a fairly decent sum of money to a publisher to make a paper open access to all L of tend journals will restrict access to paying customers if you are member of an academic Library you can get access but if you're not then you don't have access and that's one of the things about like places like the active inference Institute or the orthogonal lab is not everyone has that academic Library access and so it's really hard to get things get access to papers and things that you need so this becomes a very important set of points here gold Open Access then of course is is beneficial to the Publishers but not to the authors diamond platinum Open Access is more beneficial to the authors there's no APC fee but someone has to take that fee and you know they have to absorb that fee so that's not optimal either so this elusive zone is a way to sort of build on sort of the lowcost but still like sufficient resources we need to provide access to our work we need to open it up give everyone access to it and allow them to interact with the work in a way we couldn't even do with any of these other types of Open Access so the first thing we have is green Open Access which is pre-printing so this is where it's like not the gold standard but it's also you know you can post a paper on a preprint server there's a pre-print server that it's funded through like a foundation and you know that's that makes it open Endeavor um they're you know formal sort of you know there's a DOI which is a formal link it's a permanent link so your work is in a place that has you know some level of reputation with with a mark of approval on it you can also take that preprint and go through what's called open peer review where you open up your paper to peer review not to like three referees you've never met but to the entire community and the thing about thing about peer review and I won't get into this uh any more than what I'll talk about here but if we do peer review with like three people on an editor you know you pick the people who are maybe the most available or maybe the most visible people in the field or maybe people with the most time and they have certain opinions about what you should do what you should add on to a paper to it

publishable and sometimes it's not in the best interest of the science furthermore given that they still don't really have a very good handle on detecting things like mistakes or even fraud so you can have a paper that goes through three reviewers and an editor and still be retracted for mistakes or fraud so that's not good either open peer review allows us to have people many eyes on the paper many comments you can incorporate the comments by generating a new version of the preprint and you have this continual feedback of your work and you can eventually maybe publish it somewhere but you can also update the pre-print continually and have that process going um so that's you know Green Open Access and open peer review now a lot of papers also have data and we can include open data management in this elusive Zone because we also want to have a strategy for managing and sharing data and this goes along with the peer review and the access of the preprint that that's the link to it and that the data can then be reviewed and worked with independently of the preprint but also you know it's transparent and people can actually work with it far beyond the LIF span of a single paper then we have overlay journals so we can have preprints we can have but that's it's on a pre-print server we can have mulp versions but sometimes we want to build our own journals because we want to narrow the focus of the preprints funnel them down into things of interest for certain Fields so say for example we had an active inference overl journal it would take preprints in the active inference area only of those pre-prints and present them as Journal content so what journals do besides conferring status on people is they give you this curation of the literature of of research so that's what overlay journals can do but without the overhead of a journal so you can have like your preprints you put them on this with the skin over the top that gives us this emperor of uh you know what a journal would give you so that's the elusive zone of Open Access um and I just wanted to point that out because I think it's you know a good way to think about sort of the production of Science and the production of academic work I posted a blog post for Open Access week 2024 and it's kind of a critique of Open Access as it exists and this is why I kind of went through this and basically I argu in this post that open access is essentially failed at least up to this point that we don't have the right incentive structures we don't have the right intrinsic and extrinsic incentives to do open science correctly Open Access and so we need to change those and so you can read that post on my blog synthetic daisies so in that blog post I talk about the intrinsic and extrinsic motivators for doing Open Access sustainable so this you think of this more generally as a reinforcement learning of working open or reinforcement learning of so when I talk about intrinsic motivators I talk about the internal drive to do something so you know the idea is that

you would do things that are meaningful to yourself that's an intrinsic motivator or what motivates me in terms of a scientific question what motivates me in terms of how I can contribute to the field and then how can I benefit from or contribute to a team these are all intrinsic drives these are things that sort of you would generate and you interact with a group based on those things by contrast extrinsic motivation is the external encouragement to do something so this could be money paying people this could be cost and benefits so if there's a cost to opening my data set I do it if there's a benefit to opening my data set I will do it and that leads to an inequality within the scientific community and then of course Prestige which is of course if something is prestigious I'll do it but if something isn't prestigious I may not do it so it interacts with intrinsic motivation but it also is this external force that that sort of is encouraging people to like a you know a reinforcement mechanism extrinsic motivations can also be where things tie into existing activities so one example is where a lot of funding agencies and journals will mandate that people share their data if you want to publish a paper you have to share your data so this is like tying data sharing and doing this to publishing the problem is there's no intrinsic motivation to do this it's not meaningful to everyone and so people will do it to differing degrees of completeness or of usefulness so if you're forcing people to do something versus it's something that people want to do and think is a good idea it becomes uneven and you don't have the same quality everywhere sometimes there's you know of course there other types of mandates and other T and there's been General uh an overemphasis I think on mandates and sort of top down motivation to do things in open science and Open Access um trying to get people to adhere to certain standards or trying to enforce more uh uh trying to enforce Moray that may or may not fit the situation so this is a problem in open science Open Access and even open source where where people you know you have to kind of figure out the psychology of the people doing and enabling a lot of these open practices so if we look at this graph we see that their intrinsic motivations and extrinsic motivations that interact so at the middle of this in the gray box you have intrinsic motivations things like I want to do science I have Theory building goals I need to create tools and these things feed into extrinsic motivations such as funding Community Building uh adhering to the standards of the field and adhering to rigor standards of rigor so this is all kind of these these intrinsic and extrinsic motivations interact and then contribute to how you carry out your open science Open Access and eventually open science so I'm going to shift to sort of how we approach this in open source and so this paper from neuron it was published in 2017 it's called the commitment to open science and neur open source it's called a commitment to open source

and Neuroscience and uh P gon Andrew Davidson R Angus silver and Georgio ESC were the authors they're involved in an organization called incf which is the international neuro informatics Corp and they've all kind of done work in this area and they're contributing to a project I'll talk about in a little bit um that's open source sort of solves of problems in Neuro Imaging uh and how to do that so one thing that they point out in this paper is that code sharing is useful for a number of things in science but they make a claim that leasing code can be used to replicate results so one reason why we do code sharing is so we can replicate our results we can be confident that those results can be replicated because we have the code and everything is you know perfect um would which is not always the case but this is the claim this is the motivator that they're talking about and so sometimes you have to be strategic about how you release your code sometimes you know you just release your code as a matter of you know requirement but sometimes you can actually benefit from releasing your code and in this CA in the paper they talk about releasing core scripts and libraries before publication it will offer early feedback from the community so you can release your code just as a matter of sort of requirement or you can benefit from it by releasing your scripts and libraries early so people can debug them and help you if there are any problems with them or whatever and so you end up with this nice um optimal situation another paper that I'd like to highlight here is this paper from nature technology a technology feature in the journal Nature about six tips going public with your lab software so you know it's not enough to just write the best quality programs and to sort of put it out there you need to really kind of think about a lot of other things going on under the hood so one of those things is that you need to sort of have carve out sometime for maintenance so if you put software out into the world the reason you're one of the reasons you're doing so is to sort of serve your community you know sometimes you want to be the standard in the community other times you want to give this GI this to other researchers so that they do better research sometimes you want The Prestige of having like the top software in your field but whatever the reason you want to be able to maintain the software so that it has a longevity and so different sources that they talk about in this paper estimate anywhere from 5 to 15 hours a week to 20 30 20 to 30% of the time of the project to be spent maintaining the software now as you might guess you know there isn't a lot of motivation for that unless you know your software is really successful and so it's it can be a problem if you you know take the leap into putting your software out there but don't devote the time to maintenance it ends up hurting you later another problem is that you need to simplify runtime so a lot of times you'll release software and people will try to install it and

fail because don't have the right dependencies they don't know how to run the software and especially with a lot of academic software we kind of build a clu of plugins and other types of software on top of software so that's not ideal and that's a problem that can't be solved by necessarily by in incentives and motivations some of it is going to depend on how we implement the software and I'll talk about that in a little bit we also made want to spend time in interface design automated testing and documentation to help users who you know when you're in your own research group you're sort of in a bubble where you understand what you're doing you've used the software you understand the specific problem for which your software is written well when you release it into the public you might get people who are not as experienced or simply don't understand you know what some of the variables are or whatever and so there are a lot of ways that you know make this easy to use is a is an imperative and then a need for transparency in community building so if you really software you need to have users to make it worth your while and if you have users you want to build a community around that so that you can do things like have a more you know have a user base but also have a community where you know you can solve problems collectively and address open scientific problems and really for everyone to benefit but also transparency because we need to understand everyone's methods and we don't want to make it you know obtuse for people when they use the software to understand the methods that went into generating the result that they're get so you need to think about all these things when you open up your software when you go open source with academic software especially this brings us to another point which which is the need for research software Engineers so in Academia we usually homebrew software we end up putting it out there and you know we maybe find time to maintain it we find time to build a community but the software is not up to the standards of say professionally designed software and so one of the things we don't have is our research software engineering skills in Academia a lot of academics know the technicals matter but they don't know how to build software so software needs to be maintained and needs to undergo quality control and so those things are you know separate skills so the problem of course is that in Academia we tend to have graduate students or open source contributors build packages or whole software platforms and their contributions are well appreciated but their involvement is itinerant so that means that they might build something and then leave the lab in two years where they may not you know contribute to an open source project and you may never see them again and so they built something with no documentation and obviously no maintenance and it just kind of sits there and Withers so I've seen this a lot in my career where people will build software for one

purpose one use or like for a single paper and then it goes away because it's not maintained I mean near the water of coat I've written when I was a graduate student say where that was the case and I wish that we could recover that software and maintained it but you know who has the time so really what you need to do is you need to include research software engineer roles in your organization in your academic organization and sometimes open- Source projects will have this expertise but sometimes they don't because there's no requirement for it but it's also a different skill set than say like the academic skill set that people building the software might have so this is something where you can either cross Trin people and there programs where you can cross Trin research software Engineers but more often than not you just have to help people acquire those skills and so there are all sorts of roles for RSC in different you know academic and open- Source organizations things like intentional software design continuous integration release Cycles bug fixes and maintenance strategies and those are all things that are separate skills from the academic world separate skills even from like you know if you even if you're computer science researcher you're not necessarily a software engineer so you have to have that set of skills or at least you know have a way for people to acquire them okay now we talk about open data so open data is also a part of open science and as well as a part of Open Access so Wikipedia defines open data as openly accessible exploitable editable and shared by anyone for any purpose so it's a data set that you can access you can exploit or change you know you do analyses on or whatever you can edit it and you can share it and so anyone should be able to do this now when they say any purpose they mean that it's licensed under an open license so an open license is a permissive type of ability protection and copyright for people who put these things out there that they work on and they want to benefit from in some way but they also want others to benefit from as well so it's a legal regime to maintain you know to manage liability but also to manage the social relationship between the person making the data set producing the data set and the person using it open data allows for data to be easily shared and reused but requires govern issues of ownership and adherence to community standards so this you know part of this is you know open data is in sort of producing data sets for public consumption the other part is working on what they call an open data stack so an open data stack is a management strategy that is needed for sharing reusing And archiving data so we typically start at the bottom with some sort of interactive interface for uh working with data we need a metadata format we need a Cloud Server to host our files on to host our data and we also need standards so we can have these higher level standards for data production for open uh science production in general things need to be findable accessible

interoperable and reusable so we need to have those four principles embedded in our workflow and then we need to have a strategy for archiving data and for that we might use something we dryad which is a a data repository so we have these techn this technology layer a standards layer called security access layer and an archival layer so put together that's called a data stack and you want to build a data stack that's you know with tools that you know how to use and standards that you know how to adhere to but also things that will last you don't want to pick something that will last a year you have to migrate to something else so you want a sustainable data stack with sustainable tools with things that are easy for your team to use and so this also holds true for our Open Access publishing strategy we don't want to pick a strategy that isn't sustainable because we'll stop doing it now in the SE elegans community and this involves my work with the open roomm foundation you're fortunate in that you have many open databases so the neemod celegans is often used in biomedical research and there's some basic research on it as well and it's a tight-knit community where there are many many open data sets meaning people produce data they share them uh with the community and they have tools that are curating a lot of these data sets and papers and and things like that so we have all sorts of data sets uh available for all sorts of modes of inves a Sean's biology and behavior so we have a number of open databases covering Anatomy genetics gene expression regulatory elements Behavior conomics so this includes like the worm wearing site which includes data for onod conomics so we know in seans we know all of the neurons in the conecto we know a lot of their connections either through Gap Junctions or through synapses and we have worm wiring diagrams for both the male and the hermaphrodite so we have very rich data on worm brains we also have worm Atlas which is a atlas of structural Anatomy so on top of the connectone data you can fit structural Anatomy data and that's at worm atlas.org the seans database uh run by MDC in Berlin features a lot of data on proteomics and uh RNA okay and so this is another database that we have available senen is a database of gene expression and cgans in the nervous system so we can you know build a model of the connectome we can have the worm wiring data set senden data the worm Atlas the mdcc elegans database and we can actually you know produce a study from that so we have all those layers of data that we can include in and say a model or in an analysis we we also have this uh resource worm base which is where we can explore the world of seans and find mutants so seans has a number of defined mutants for specific genes they specific mutations that we can backcross and so if we have a strain of say some mutant defin mutant we have this worm that has a certain mutation that might be expressed in some Behavioral or anatomical deficit and so we can compare that to the wild

type and say things about mutations of function and so this this is a really useful resource both for experimental work and for looking at the celan's genome and finding you know like alignments and other types of data like in one of papers we recently did we found uh different types of we aligned different uh open reading frames within gen so it's really a good resource and then of course uh nature methods this paper uh by people in the open worm Foundation um and this is called an open source platform for analyzing and sharing worm Behavior data and this is um on something called the turpy tracker which is a movement tracker so they can actually take microscopy images of celegans they can track the worms movement and they can characterize different movements they they're stereotypical movements worms make and can actually track the behavior of the worms and classify so that's you know that that's the kind of data that's available in celegans and so if you think about the active influence Community you know what kinds of data sets and what kinds of resources could be built around things involving active inference so you might for example think about you know and some of these resources are just more General resources that you could bring to bear so you might say for example you know go to a neuro Imaging resource and then go to a behavioral resource or build a behavioral resource there all sorts of ways that you can build these um open data repositories to help the community Aid the community in solving problems and answering questions so this is a talk by ecof freed this is called open science is an antidote to poor practice cynicism so one of the things he advocates for on this video video is to use open science as a way to combat poor practice like we said open science is about reinforcing practice and when you reinforce practice of open science you reinforce good scientific practice in general because you're focusing on the process you're focusing on you know cross checks and and you know other things that involve teamwork and so there there's a lot of this sort of advocacy for helping science through open science but also combating cynicism where when we see open when we see poor science we often think well this is just the state of science that it's actually failing or it's bad and so you can fight disinformation as well with open science this is a nice set of views here so in the orthogonal lab we have done a lot of work modeling open source communities and their sustainability so I talked earlier about Open Access sustainability now I'm going to talk about open source sustainability and so we've done a lot of work with modeling Frameworks to understand this problem so why is sustainability important well in the case of Open Source many projects fail for a host of reasons sometimes they're related to people's projects you know not being of good scope where they weren't asking very good questions sometimes the team falls apart the team is dysfunctional

sometimes the project runs out of money so there are a lot of reasons why projects fail and sometimes open source projects especially have problems with maintainers problems with people contributing so they fail for those reasons as well so you can have things like we're scoping of a project we're trying to do too many things we're trying to do too few things we might have bad management or a lack of participation or we might have no resources in terms of time money so these are all things that we want to be able to figure out how to manage at a reasonable level and to do that we have to understand sustainability so in these you know in these models that we've built we use different methods including active inference reinforcement learning and large language models to model open source communities and understand what makes a project sustainable so the first one is this pre-print on that kind of includes a lot of the work that we've done uh especially in terms of reinforcement learning models and active influence models this is our research gate and then there's this new work on com combining agent based models and large language models this is Sarah Bostawala's Google Summer of Code project from this past year so she actually presented in front of the Google Summer of Code um uh Symposium and and she presented her project there and so this work so this work on large language models involved using large language models to generate issues and then generate agents that could solve those issues and see which kinds of you know almost create fake projects and then create issues to solve and then see how the agents were able to solve them and so you know we use the combination of agent based techniques and large language model techniques to sort of work this a very exciting project another approach is to take the multi-agent reinforcement learning route and so you see this grid down here and this figure and all these dots and they represent agents that are doing different things sometimes they have different statuses open source Community sometimes they're of a different level of Competency and you can actually run these agent based models you can Implement them in Misa as a python package and run these agent based models and then use reinforcement learning algorithms to sort of optimize the strategies that they're using so this is you know again we are using these uh sets you're creating these projects you're using these sets of issues and you're measuring their ability or their competence to solve those problems and so uh sh pomon RV Roger gopalan and hu shogo were all involved in this work over the past several years we've been doing these kind of reinforcement learning models for understanding open source syst sustainability then we've uh done this work on Collective cognition for AI ethics this is where I actually applied active inference and we modeled agent behavior in terms of active inference using a package called interactively and this is the GitHub repo here

interactively this applies a partially observable partially observe markov decision process or a pomdp model uh to the problem we also used a multi arm banded approach to model explore and exploit behaviors so explore and exploit is sort of this tradeoff that agents make between exploring a project exploring issues to be solved and exploiting specific issues solving them so it's like making you know coordinating agent behaviors and then figuring out what strategy they're going to use and modeling all of that so very interesting project uh it needs more work on it so if people in this group in in the active inference Institute are interested and following up on it that would be welcome and so there are a number of issues about maintenance that are very important to sustainability and this is a special in the academic sphere because academic organizations and open source typically don't have the resources that private organizations might so maintenance assist sustainability issues are multitude in open source so I'll talk about a couple of them here so the first one is log4j and this happened a couple years ago where there were a number of Open Source projects that got hit with the security threat and it was because their security fell at a sync with the current state-of-the-art so in open source we don't maintain things as well as say commercial products so you have these holes that can be exploited by malicious actors so log4j was so serious that they had to um consult with the US government to try to help fix this problem and so this was a big problem um with just kind of maintaining the security of software but also it exposed a big hole in the open source model and so this is still something that probably requires meta governance you know some organization between like the federal government and open source organizations to solve the second is called the make or taker problem which is basically where some organizations make open source software and other organizations fork or copy the software for their own purposes thus being a taker and the problem here is where that relationship is not reciprocal that you know the makers will develop systems that encourage forks and then the open source contributors or the takers are expected to contribute back to the makers some sort of you know maintenance or something like that but they failed to do so and so this is the problem where you need to coordinate this reciprocal relationship and so an example of this is the word press for controversy so WordPress is the maker in this case and like I think it's WP engine is the taker in this case WP engine was forked from more press and it resulted in a fight between the two organizations one being the you know the maker and one being the fork or The Taker and they've been fighting over commercial issues and fair use issues that have served to break the Norms of Open Source so there's this primary issue of maintaining sort of a reciprocal relationship between organizations that rely on basically rely on each other to survive but you

can have all sorts of you know fights and disagreements about different ways that the software can be used commercially and other types of use that you know are although are stated in the license sometimes evolve away from the license and their practical concerns that you know isn't going to be solved by just some sort of reciprocal relationship or even intrinsic and in extrinsic motivations and then finally the final example is XZ back door SL auto tools this was an exploit that was made in in an open source repository where it was it resulted from a lack of maintainer vigilance so this maintainer in this particular case was burnt out and they didn't feel like really kind of closely inspecting every pull request which is where a contributor will submit a change or contribution to an open source organization and so ious malicious actor was able to get exploits into the software and it became a huge security problem it's kind of like L forj and the problem here is it's you know the failure here is at the level of maintainer and it wasn't the maintainer fault necessarily it's that sometimes people in open source organizations as well as Open Access organizations open sites have become burnt out especially when you have to do all these things all these practice oriented things to realize open vision and you know also you have disincentives that lead to problem so if you're not paying your maintainers enough or if you're not Mo you know if they don't have that intrinsic motivation high enough they can't do the prop the job that they need to do to make things work or the extrinsic motivations aren't well good enough you don't really pay or maintain her very much and therefore they're disincentivized to sniff out exploits and so these are our problems with maintaining open source in an especially in an academic environment this is a second uh YouTube video I wanted to point out this is behind the scene behind the screen there's a word play here which is open source software is both a tool and topic for research this is from William Schuler at Santa Fe Institute and he talks about sort of the history of software and how you know you can research open source software but also use it as a tool it has a lot of interesting issues in here so I recommend watching this video now I'm going to talk about how to sort of implement open software for like a purpose or an organization so we talked about you know one of these points about Distributing open academic software is being being able to run it properly or run it in a way that doesn't rely on too many dependencies and so this is where we can use something like Linux which or you know some flavor of Linux to package data analysis tools so that people can use them in an environment where they don't have to worry about dependencies they don't have to worry about having the proper software requirements so there are two examples of where we can use a Linux dist to package data analysis tools and present them to people in the community sometimes with fairly low

technical standards so the first is narof Fedora and so this was created as free and open source software this is where we have a version of the Fedora this Dr of Linux and we in Fedora you package a bunch of neuroimaging analysis tools and you put them in and you just give people the package they install it on their machine and then they can run those tools without worrying about dependencies or getting thing onun on Windows or Mac or whatever the other example is neurod Debian this is run by incf this is the group led by por gleon and and Angus or Angus um run by run by or Leon and and colleagues and this is where they have a duby and drro of Linux and they have these neuroimaging tools in Debbie and so you have Fedora and Debian you can choose between the two dros and you can run these enging Tools in these environments and that's that makes it easier I think for neuroscientists to do this kind of work to work with these packages uh similar thing could be done for active inference and especially in Fedora you have these different flavors of Fedora like for artists or for musicians where you can package tools for those professions and they can just you know use it out of the box so focusing on the neurop Fedora uh this is a special interest group whose goal is to provide Neuroscience researchers and enthusiasts with a strong easy to use readymade free and open source software platform for their work and anchor Sinha has let the taken the lead under aladora Morgan Huff is part of the orthogonal lab has also worked on this as well then we have neurodes so neurodes is not an operating system but it's a virtual machine it's a virtualization or a containerized environment where you have like basically virtual machine on your existing computer and you run everything inside of that and it really paves over a lot of the operating system differences and system requirement differences that you might find trying to install open source software on your home computer so like you know this is where you've basically created a controlled environment for the software so it really helps people run it when they have very little Tech technical expertise and just get it to run that's all you need to do so neurodes uh is a containerized data analysis environment specialized for neuroimaging research and this is something that you could easily do with the active inference it's where you package the software code with the operating system libraries and dependencies required to run the code in a single lightweight executable so this is platform agnostic it's just you open up the container you run it and you can do everything inside of there so thank you for your attention um I especially want to thank the members of the open source interest group where Bradley Alisa Jesse parent Morgan Huff Sarah Bala Brian mccorkel amanu shle rvy Roger gopalan and Shu pamon um we have other people who cycle through the group from time to time and you're welcome to join us uh please join the worth going a research lab and then be

on the lookout for open- Source activities we usually do the interest group in the summer if
you're interested get in touch otherwise Happy active inferring